

Carrier Hierarchy ABI v0

Predicate, Linear, Matrix, Algebraic, Projective, and Atomic Layers

Status: Normative Draft

Scope: Carrier hierarchy, barcode transport, frame preservation, validation ordering

Depends on: `omi.delta_orbital_lexer.v0`

Canonical frame: `S = uint16_t[8]`

Canonical invariant: `decode(encode(S)) = S`

Canonical validator: `Q(S) = 0`

1. Purpose

This document defines the ordered carrier hierarchy for transporting the canonical Δ_C -orbit frame:

```
S = uint16_t[8]
```

with canonical segment layout:

```
0xLL00-0x03bf-0xNNNN-0x2bLL-0x2fLL-0xMMMM-0x039f-0xLLff
```

The carrier hierarchy does not define new semantics. It defines how physical, linear, and matrix barcode surfaces preserve the same canonical frame before algebraic validation.

The central rule is:

```
The algebraic frame is constant.
The carrier surface is variable.
The proof does not change.
Only the transport surface changes.
```

2. Complete Processing Chain

The full carrier-to-execution chain is:

```
Physical:  ITF → Code39
Transport:  Codabar → Code128 → Code16K → Code93
```

Geometric: Aztec → MaxiCode → JABCode → BeeTag
Algebraic: $Q(S) = 0$
Projective: δ_C_LL resolver
Atomic: CAS cursor

Expanded:

```
physical signal
  ↓
predicate gate
  ↓
linear precision ladder
  ↓
matrix projection plane
  ↓
canonical S = uint16_t[8]
  ↓
Q(S) = 0
  ↓
extract LL, NN, MM
  ↓
 $\delta\_C\_LL$  truth-row resolution
  ↓
steps ∈ valid bound
  ↓
CAS cursor advance
  ↓
64-bit receipt write
```

No downstream layer may execute before the previous authority condition is satisfied.

3. Predicate Gate

The predicate gate establishes whether a scan or pulse sequence is eligible for segment extraction.

It consists of:

ITF → clock predicate
Code39 → guard predicate

3.1 ITF Clock Predicate

ITF answers:

Is there a valid pulse boundary?

Its role is physical synchronization:

clock recovery
pulse boundary detection
metric timing
scan rhythm
start-of-signal eligibility

ITF does not validate the frame. It only establishes that a physical pulse stream is coherent enough to continue.

3.2 Code39 Guard Predicate

Code39 answers:

Are the start/stop guards present?

Its role is guard confirmation:

start guard
stop guard
gross scan boundary
symbol eligibility

Code39 does not extract the canonical truth row. It confirms that the carrier has a guarded symbolic boundary.

3.3 Predicate Gate Rule

Both predicates must pass before the linear ladder begins:

$\text{ITF_CLOCK_VALID} \wedge \text{CODE39_GUARD_VALID}$

If either predicate fails, the scan is rejected before segment extraction.

4. Linear Ladder

The linear ladder is the one-dimensional precision and redundancy ladder.

It consists of:

Codabar → Code128 → Code16K → Code93

Each step increases transport precision or redundancy.

Layer	Carrier	Role	Precision Function
1	Codabar	raw carrier	binary16 segment extraction
2	Code128	compact carrier	binary128 checksummed frame
3	Code16K	multi-row carrier	binary192 expanded transport
4	Code93	witness carrier	binary256 error-corrected witness

5. Codabar Raw Segment Layer

Codabar is the raw segment extraction layer.

It answers:

Can the physical carrier yield 16-bit segment material?

Role:

raw segment transport
simple delimiter mapping
low-cost scan carrier
binary16 extraction

Codabar is closest to the raw `uint16_t` segment layer. It may carry direct segment material or a sequence that reconstructs segment material.

The output of this stage must be reducible to:

```
S[0..7]
```

or to partial material that a higher ladder stage verifies.

6. Code128 Compact Frame Layer

Code128 is the compact checked frame carrier.

It answers:

```
Can the 128-bit frame be compactly transported with a carrier checksum?
```

Role:

```
binary128 frame transport  
dense one-dimensional encoding  
checksum-backed segment preservation  
compact printed form
```

Code128 may carry the entire canonical frame:

```
S = uint16_t[8]
```

Carrier-level checksum success does not imply algebraic validity. It only means the Code128 symbol decoded according to Code128 rules.

The frame must still pass:

```
Q(S) = 0
```

7. Code16K Multi-Row Layer

Code16K is the multi-row expansion layer.

It answers:

Can the frame and additional transport material span multiple rows?

Role:

binary192 transport profile
multi-row sequence support
larger body transport
frame plus auxiliary material

Code16K may carry:

full frame
frame plus checksum material
frame plus receipt material
frame plus redundant row data

It does not change the frame authority model. Any recovered frame still reduces to `S = uint16_t[8]`.

8. Code93 Witness Layer

Code93 is the highest one-dimensional witness layer.

It answers:

Can the carrier witness and verify the lower ladder?

Role:

binary256 witness carrier
stronger check-character structure
redundant frame witness
one-dimensional error-detection profile

The Code93 witness may verify material from the lower ladder:

Codabar raw extraction
Code128 compact frame
Code16K multi-row assembly

But it remains a carrier-level witness. It does not replace `Q(S)`.

The locked rule is:

Code93 may witness transport integrity.
Only `Q(S)` proves frame authority.

9. Matrix Planes

The matrix planes are two-dimensional geometric projection surfaces.

They consist of:

Aztec → MaxiCode → JABCode → BeeTag

These planes are assigned according to the C0 separator hierarchy:

FS → File / Frame
GS → Group
RS → Record
US → Unit

Plane	Carrier	Role
FS	Aztec	world frame / orientation detection
GS	MaxiCode	group carrier / batch frame transport
RS	JABCode	record carrier / high-capacity storage
US	BeeTag	unit carrier / short-link pointer

10. Aztec FS Plane

Aztec occupies the FS plane.

It answers:

Can the scan establish global orientation?

Role:

```
finder pattern  
orientation recovery  
world-frame projection  
robust 2D frame carrier
```

Aztec is appropriate when the system needs a strong visual orientation anchor.

Its output must reduce to:

```
S = uint16_t[8]
```

or to a pointer that resolves to `S`.

11. MaxiCode GS Plane

MaxiCode occupies the GS plane.

It answers:

```
Can multiple frames be grouped in one scan context?
```

Role:

```
batch grouping  
group transport  
multi-frame scan context  
logistics-style carrier
```

MaxiCode is suited for grouped frame transport where several frames or receipts need to be bound to a common carrier context.

The group does not alter the authority of individual frames. Each decoded frame must still pass independently through:

```
Q(S) = 0
```

12. JABCode RS Plane

JABCode occupies the RS plane.

It answers:

Can high-capacity record material be stored on a 2D surface?

Role:

record storage
high-capacity visual carrier
color-module transport
frame plus auxiliary data

JABCode may carry complete frames, frame sequences, receipts, or redundancy material.

Its record capacity does not change validation authority. Every recovered canonical frame must still be algebraically validated.

13. BeeTag US Plane

BeeTag occupies the US plane.

It answers:

Can a compact visual unit point to a stored frame?

Role:

unit pointer
short visual tag
compact link
low-capacity reference carrier

BeeTag is best interpreted as a pointer carrier rather than a full-frame carrier.

It may resolve to:

```
stored S frame
receipt slot
frame database key
local cache entry
```

The referenced object must eventually reduce to canonical S and pass $Q(S) = 0$.

14. Carrier Invariant

Every carrier shares the same preservation invariant:

```
decode(encode(S)) = S
```

for all supported structurally valid frames within that carrier's capacity and error-correction profile.

The carrier may add:

```
checksum
guard
finder pattern
row structure
color channel
redundancy
pointer indirection
```

But it may not add new frame semantics.

The frame is the authority.

The carrier is the surface.

15. Authority Ordering

The complete authority order is:

1. Predicate authority
ITF clock and Code39 guard
2. Transport authority
linear ladder reconstruction

3. Geometric authority
matrix orientation, grouping, storage, or pointer resolution
4. Algebraic authority
 $Q(S) = 0$
5. Projective authority
 δ_C_LL truth-row closure
6. Atomic authority
CAS cursor establishes replay order

Each layer answers a different question.

Layer	Question
ITF	Is there a valid pulse boundary?
Code39	Are guards present?
Codabar	Can raw segment material be extracted?
Code128	Can a compact checksummed frame be recovered?
Code16K	Can a multi-row body be assembled?
Code93	Can a high-redundancy witness verify transport?
Aztec	Can global orientation be recovered?
MaxiCode	Can grouped frames be transported?
JABCode	Can record material be stored and recovered?
BeeTag	Can a compact unit pointer resolve?
$Q(S)$	Is the frame structurally valid?
δ_C_LL	Does the truth row close under the bound?
CAS cursor	Is replay order atomically assigned?

16. Rejection Rules

A conforming implementation must reject as early as possible.

16.1 Predicate Rejection

Reject before segment extraction if:

```
ITF pulse boundary fails  
Code39 guard fails
```

16.2 Transport Rejection

Reject before algebraic validation if:

```
linear carrier checksum fails  
multi-row assembly fails  
witness mismatch occurs  
decoded length cannot produce S[0..7]
```

16.3 Geometric Rejection

Reject before algebraic validation if:

```
matrix orientation cannot be recovered  
group frame is incomplete  
record decode fails  
pointer target cannot be resolved
```

16.4 Algebraic Rejection

Reject before truth-row interpretation if:

```
Q(S) ≠ 0
```

16.5 Projective Rejection

Reject before cursor movement if:

```
fanoTruthResolver(LL, NN, MM) = -1
```

16.6 Atomic Rejection

Retry, not reject, if CAS loses a race:

```
compareExchange failed because another writer advanced the cursor
```

A CAS race is not a malformed frame. It is normal concurrent execution.

17. Engineering Interpretation

The hierarchy can be understood as a progressive narrowing of uncertainty.

Physical layer:

Is there a usable signal?

Guard layer:

Is the signal framed?

Linear layer:

Can the one-dimensional carrier recover segment material?

Witness layer:

Can transport integrity be checked?

Matrix layer:

Can orientation, grouping, storage, or reference be recovered?

Algebraic layer:

Does the recovered frame prove structural authority?

Projective layer:

Does the truth row resolve inside the bound?

Atomic layer:

Can the accepted row receive a unique replay position?

This is the engineering value of the hierarchy. Each layer has one job.

18. Minimal Canonical Chain

The full chain is:

ITF

- Code39
- Codabar
- Code128
- Code16K
- Code93
- Aztec
- MaxiCode

- JABCode
- BeeTag
- $Q(S) = 0$
- δ_C_LL resolver
- CAS cursor

This can be grouped as:

Predicate:

ITF → Code39

Linear:

Codabar → Code128 → Code16K → Code93

Matrix:

Aztec → MaxiCode → JABCode → BeeTag

Computation:

$Q(S) = 0$ → δ_C_LL → CAS cursor

19. Locked Statement

The carrier hierarchy is ordered as follows:

Physical: ITF → Code39
 Transport: Codabar → Code128 → Code16K → Code93
 Geometric: Aztec → MaxiCode → JABCode → BeeTag
 Algebraic: $Q(S) = 0$
 Projective: δ_C_LL resolver
 Atomic: CAS cursor

Every carrier preserves the same invariant:

$\text{decode}(\text{encode}(S)) = S$

The algebraic frame is the constant.

The carrier is the variable.

The proof does not change.

Only the surface changes.

20. Final Summary

This hierarchy separates physical signal detection, guarded carrier transport, linear precision, matrix geometry, algebraic validation, projective truth-row resolution, and atomic replay indexing.

The architecture is therefore not:

```
barcode equals program
```

It is:

```
barcode carries frame  
frame proves structure  
resolver proves relation  
cursor proves replay order
```

The carrier hierarchy exists to move the frame through physical and visual media without changing its authority.